

RISHAV RAJ

AI/ML Engineer | LLM Fine-Tuning, Evaluation & Agent Security

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Summary

M.Sc. Computer Science graduate (2026) who fine-tunes LLMs, tests them on unseen data, and works out why they fail. Took text-to-SQL execution accuracy from 50.6% to 68.4% on 21 held-out databases, and ran 100 live prompt-injection attacks against an LLM agent. Thesis on why TIES merging of distilled LoRA adapters ends up as plain averaging.

Skills

Languages & Tools: Python, SQL, Linux, Git, GitHub

LLM & ML: PyTorch, Hugging Face Transformers, PEFT/LoRA, TRL, GRPO, Unsloth, knowledge distillation, model merging (TIES), RAG, LangGraph

Evaluation & Data: execution-based evaluation, held-out and database-level splits, significance testing, adversarial testing, MLflow, embeddings, Pinecone

Serving & Cloud: vLLM, llama.cpp, GGUF quantization, FastAPI, Docker, AWS (EC2, Lambda, CloudWatch)

Projects

Text-to-SQL Fine-Tuning & Deployment | Qwen2.5-Coder-7B

Mar – Jun 2026

github.com/dLounge/Text-to-SQL | [Demo](#)

- Fine-tuned Qwen2.5-Coder-7B with LoRA SFT, then GRPO with SQL execution results as the reward. Execution accuracy went from 50.6% to 68.4% on 1,600 questions from 21 databases held out of training.
- Split the data by database so no schema appears in both training and test, and dropped examples whose gold SQL failed, timed out or returned no rows. Checked for exact duplicates, near-duplicates and overlap across splits. Built the GRPO set from prompts the SFT model solved in some, but not all, of 8 samples.
- Quantized the SFT model to GGUF Q4_K_M (15 GB to 4.4 GB, 0.6-point accuracy drop) and served it on AWS EC2 with llama.cpp and FastAPI, about 12 seconds per query on CPU. Added input checks, a timeout, health checks and CloudWatch metrics; tracked four runs in MLflow.

Sentinel Procurement | LangGraph Agent Security

Jun – Aug 2026

github.com/dLounge/sentinel-procurement

- Built a procurement agent where the model that reads untrusted vendor text has no tools, and the model that places orders sees only validated, typed offers. Budget, price and delivery checks with no model in them must pass before any order exists.
- Ran 100 live prompt-injection attacks across 10 attack families. 12 moved the buyer's decision beyond normal run-to-run variation, but none produced an unauthorized order or a policy violation.
- Compared four architecture variants in a 25-cell test matrix: the three weakened variants let unauthorized orders through, the full design did not. 310 automated tests cover the checks, the negotiation flow and the evaluation code.

KD-TIES: TIES Merging of Distilled LoRA Adapters | M.Sc. Thesis

Dec 2025 – Jun 2026

github.com/dLounge/KD-ties

- Distilled Qwen2.5-32B into LoRA adapters for Qwen2.5-7B (instruction following and math, 5,000 samples each) and merged them with TIES. For adapters trained on the base model and merged into the instruct model, sign conflict was 0.000, so TIES behaved like plain averaging.
- Traced this to the instruction-tuning shift inside every task vector. Subtracting it before merging (SC-TIES) brought sign conflict back to 0.237 and gave the only cross-method gain above 2 standard errors (+0.017 MMLU).
- Reported the negative result: distillation had already cut GSM8K from 0.632 to 0.466 before any merge, and no merge variant recovered it.

Experience

Artificial Intelligence Intern | Ballistic Learning Systems, Faridabad, India

Jun – Jul 2025

- Compared an LLM grader with 118 human-labelled answers and raised agreement from 44.1% to 71.0% by changing the prompt and the retrieved context.
- Built a RAG pipeline over course PDFs: chunked the documents, stored embeddings in Pinecone, and retrieved context for grading and question generation.

Education

M.Sc. in Computer Science

Jul 2024 – Jun 2026

Central University of Rajasthan, Ajmer | CGPA: 7.1/10

Bachelor of Computer Applications

Aug 2018 – Jul 2021

Patliputra University, Patna | 73%